REAL TIME SMART FACTORY MANAGEMENT & FAULT DETECTION USING TEMPERATURE , FLAME AND VIBRATION

A

Industry-Oriented Mini Project Report

Submitted in the Partial Fulfillment of the

Academic Requirements

for the Award of the Degree of

Bachelor of Technology

In

Electronics and Communication Engineering

By

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ACE Engineering College

(NBA ACCREDITED B. TECH COURSES ECE, EEE, CSE, MECH, CIVIL)

(NAAC" A" GRADE)

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Ankushapur(V), Ghatkesar(M), R.R.Dist - 501 301

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

CERTIFICATE

This is to certify that the Industrial Oriented Mini Project entitled **"REAL TIME SMART FACTORY MANAGEMENT AND FAULT DETECTION USING TEMPERATURE , FLAME AND VIBRATION"** done by

B. YOKSHITHA **23AG1A0403**

Department of Electronics and Communication Engineering is a record of bonafide work carried out by them. This Industry-Oriented Mini Project is done as partial fulfilment of Obtaining Bachelor of Technology degree to be awarded by **JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY, HYDERABAD,** during the academic year 2025 – 26.

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With Regards:

B. YOKSHITHA 23AG1A0403

ABSTRACT

Industrial environments require continuous monitoring to maintain safety, reliability, and productivity. Faults such as machine overheating, fire accidents, and abnormal vibrations can lead to serious damage to equipment and may also cause production losses. To prevent such problems, an efficient real-time monitoring and fault detection system is required in modern factories.

The proposed project, **Real Time Smart Factory Management and Fault Detection using Temperature, Flame, and Vibration Sensors**, is designed to monitor industrial conditions and detect faults at an early stage. The system uses sensors to measure temperature levels, detect fire through flame sensing, and monitor machine vibrations. These sensors are connected to a microcontroller which continuously reads and analyzes the sensor data.

When any abnormal condition such as high temperature, flame detection, or excessive vibration occurs, the system immediately activates safety measures such as a buzzer alarm and water sprinkler through a relay module. Additionally, the NodeMCU module enables IoT connectivity, allowing real-time monitoring and remote access to system data through the internet.

The system helps industries improve operational safety by providing early warnings before major failures occur. It also reduces the need for constant manual supervision and allows factory operators to respond quickly to emergency situations. By integrating sensor technology with IoT communication, the proposed system offers an effective solution for implementing smart monitoring and fault detection in modern industrial environments.

CONTENTS

CHAPTER 1
INTRODUCTION

1.1 Introduction of the project

Modern industries are rapidly adopting automation technologies to improve productivity, safety, and efficiency in manufacturing environments. A factory consists of several machines and electrical systems that operate continuously for long periods of time. During operation, machines may experience different types of faults such as overheating, fire hazards, or abnormal vibrations. These faults may lead to equipment damage, production loss, and sometimes dangerous accidents for workers. Therefore, continuous monitoring of industrial equipment and surrounding environmental conditions is very important to maintain safety and smooth operation of the factory.

Traditional industrial monitoring methods mainly depend on manual supervision and periodic maintenance. In these methods, workers inspect machines and factory conditions at regular intervals to detect possible faults. However, this approach is not always reliable because faults may occur suddenly between inspection periods. If such problems are not detected early, they can lead to serious damage to machinery and may even cause fire accidents or system failures. Because of these limitations, industries are now moving toward smart monitoring systems that can automatically detect faults and provide real-time alerts.

The concept of a smart factory is based on the integration of sensors, microcontrollers, and communication technologies to create an intelligent monitoring system. Sensors are used to measure different physical parameters such as temperature, flame detection, and vibration levels. These sensors continuously collect real-time data from machines and the surrounding environment. The collected data is processed by a microcontroller which analyzes the information and determines whether the system is operating under normal conditions or if any abnormal situation has occurred. The proposed system uses temperature, flame, and vibration sensors to monitor industrial conditions and detect faults at an early stage. The sensors are connected to a microcontroller which processes the data and activates warning systems such as buzzers and water sprinklers when abnormal conditions occur. In addition, the system uses NodeMCU to provide IoT connectivity so that the factory environment can be monitored remotely. This system improves industrial safety, reduces manual monitoring, and helps industries maintain efficient factory operations. Furthermore, the implementation of such smart monitoring systems supports the development of modern smart factories where machines, sensors, and communication technologies work together to improve efficiency and safety. By continuously monitoring critical parameters like temperature, fire presence, and vibration levels, industries can prevent major failures and reduce maintenance costs. This system also helps in improving reliability and ensures uninterrupted industrial operations.[5]

1.2 LITERATURE SURVEY

Industrial monitoring and fault detection systems have gained significant attention in recent years due to the rapid growth of automation and smart manufacturing technologies. Researchers and engineers have developed various monitoring systems to improve safety and efficiency in industrial environments. These systems mainly focus on detecting abnormal conditions in machines and factory surroundings to prevent equipment damage and ensure smooth industrial operations.

Several studies have focused on temperature monitoring systems to detect overheating in machines and electrical equipment. Excessive temperature in industrial machines may lead to component failure, reduced performance, and even fire hazards. Temperature sensors are widely used in industrial applications to continuously monitor heat levels and alert operators when the temperature exceeds safe limits. By implementing temperature monitoring systems, industries can prevent major equipment damage and maintain operational stability.

Flame detection systems have also been widely researched and implemented for industrial safety applications. Fire accidents are one of the most dangerous risks in factories and manufacturing plants. Flame sensors can detect infrared radiation produced by fire and provide early warnings before the fire spreads. Many researchers have developed automated fire detection systems that activate alarms or fire suppression systems immediately when a flame is detected, helping to reduce the risk of serious accidents.

Another important area of research in industrial monitoring is vibration analysis. Machines in factories often produce vibrations during operation, but abnormal vibration levels may indicate mechanical problems such as imbalance, misalignment, loose components, or bearing wear. Vibration sensors are used to monitor machine stability and detect unusual vibration patterns. By analyzing vibration data, industries can perform predictive maintenance and identify faults before they lead to machine breakdown.

With the advancement of Internet of Things (IoT) technology, many modern industrial monitoring systems are designed to transmit sensor data through wireless networks. Microcontrollers such as Arduino and communication modules such as NodeMCU or ESP-based systems are commonly used for implementing IoT-based monitoring systems. These systems allow real-time data collection, remote monitoring, and faster response to abnormal situations.

The integration of multiple sensors in a single monitoring system has proven to be more effective in detecting various types of industrial faults. By combining temperature sensors, flame sensors, and vibration sensors, industries can monitor different environmental and machine conditions simultaneously. Such integrated systems improve industrial safety, reduce the chances of unexpected failures, and support the development of smart factory environments.

1.2 Proposed System:

The proposed system is designed to develop a real-time smart factory management and fault detection system that monitors important industrial parameters such as temperature, flame detection, and machine vibration. In modern industries, faults like overheating, fire accidents, and abnormal vibrations can cause serious damage to machines and may interrupt the production process. Therefore, the proposed system aims to continuously monitor these parameters and detect faults at an early stage to ensure safe and efficient factory operations.

In this system, different sensors are used to monitor the working conditions of the factory environment and machinery. A temperature sensor is used to measure the temperature levels around machines and detect overheating conditions. A flame sensor is used to identify the presence of fire or flame in the factory environment. A vibration sensor is used to monitor machine vibrations and detect abnormal vibration levels that may indicate mechanical faults or instability in machines.

All these sensors are connected to a microcontroller which acts as the central processing unit of the system. The microcontroller continuously receives data from the sensors and compares the sensor values with predefined threshold levels. If any parameter exceeds the safe limit, the system immediately identifies it as a fault condition. This allows the system to respond quickly to abnormal situations and reduce the chances of major industrial accidents.

When a fault is detected, the system activates warning and safety mechanisms. A buzzer is used to provide an audible alert so that workers and operators in the factory are immediately informed about the problem. In case of fire detection, a relay module is used to activate a water sprinkler system which helps in controlling the fire and preventing further damage to the equipment and surrounding environment.

To enhance the monitoring capability of the system, a NodeMCU module is integrated to provide Internet of Things (IoT) connectivity. This allows sensor data to be transmitted through Wi-Fi so that factory managers and operators can monitor the system remotely using internet-based platforms. Real-time monitoring enables quick decision making and helps in improving overall factory safety and management.

The proposed system is simple, cost-effective, and efficient for industrial monitoring applications. By combining multiple sensors, microcontrollers, and IoT communication, the system provides continuous monitoring of factory conditions and ensures early detection of faults. This helps industries reduce equipment damage, prevent accidents, and maintain smooth and reliable industrial operations.

CHAPTER 2
HARDWARE AND SOFTWARE DESCRIPTION

2.1 HARDWARE COMPONENTS

Hardware components play an important role in the implementation of the proposed smart factory monitoring and fault detection system. These components are responsible for sensing environmental conditions, processing data, and performing necessary actions when abnormal conditions occur. In this project, different sensors such as temperature sensor, flame sensor, and vibration sensor are used to monitor factory conditions. These sensors collect real-time data and send it to the Arduino UNO microcontroller for processing.

The Arduino UNO acts as the main control unit of the system. It receives signals from the sensors and analyzes the data to determine whether the system is operating under normal conditions. If the sensor values exceed the predefined limits, the Arduino activates output devices such as a buzzer and water sprinkler through a relay module. The NodeMCU module is used to provide IoT connectivity so that the sensor data can be transmitted through Wi-Fi for remote monitoring.

Other hardware components such as the buzzer, relay module, and water sprinkler are used for alert and safety mechanisms. The buzzer provides an audible warning signal when any fault condition is detected in the factory environment. The relay module acts as an electronic switch that controls highpower devices like the water sprinkler system. The water sprinkler helps to control fire accidents automatically when a flame is detected.[6]

By integrating all these hardware components, the proposed system can effectively monitor industrial conditions and respond quickly to dangerous situations. The use of sensors, microcontrollers, and IoT communication modules makes the system reliable, efficient, and suitable for smart factory environments.

2.1.1 Arduino UNO Overview

Arduino UNO is one of the most widely used microcontroller development boards in embedded system applications. It is based on the ATmega328P microcontroller and is designed to make electronics and programming easier for beginners, students, and professionals. Arduino UNO provides a simple and flexible platform for developing various automation and monitoring systems. Because of its easy programming environment and wide range of applications, it has become very popular in educational institutions and industrial prototype development.

The Arduino UNO board contains both digital and analog input/output pins which allow it to interact with different sensors and electronic devices. These pins help the microcontroller to receive signals from sensors and send control signals to output devices. The board can be easily programmed using the Arduino Integrated Development Environment (IDE), which supports programming languages based on C and C++. This makes the development process simple and efficient.

Another advantage of Arduino UNO is its compatibility with a large number of external modules and sensors. Various components such as temperature sensors, flame sensors, vibration sensors, relays, and communication modules can be connected easily to the board. This flexibility allows developers to design complex monitoring and automation systems without much difficulty.

In this project, the Arduino UNO acts as the central control unit of the smart factory monitoring system. It receives signals from the temperature, flame, and vibration sensors and processes the data continuously. If the sensor values exceed the predefined safety limits, the Arduino activates warning devices such as a buzzer and controls the relay module to operate the water sprinkler system.[1]

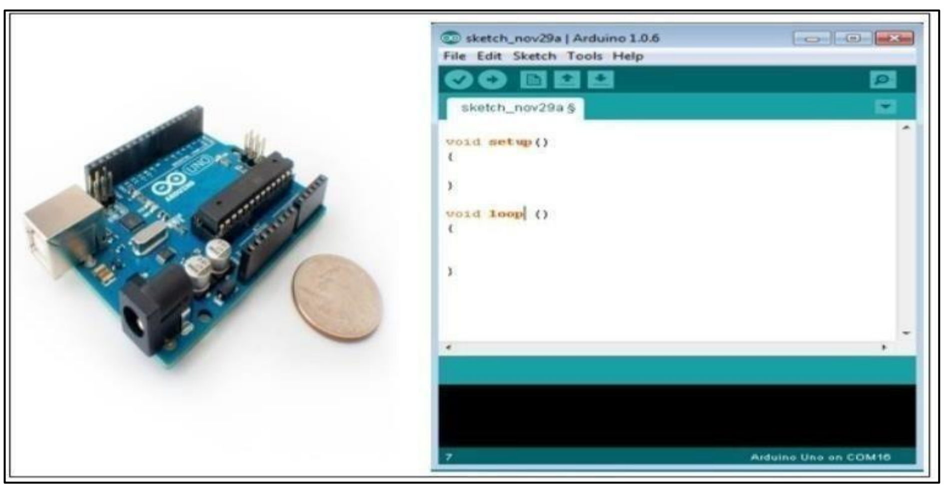


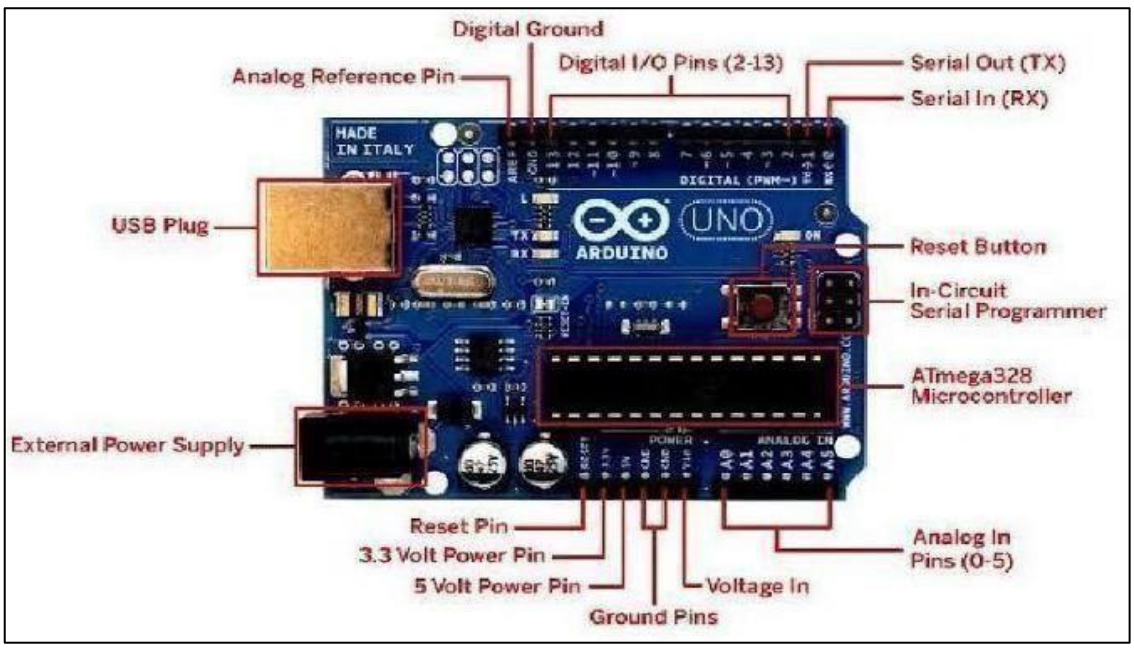
Fig 2.1.1 Arduino Uno

2.1.2 Arduino UNO Description

The Arduino UNO board is built around the ATmega328P microcontroller which is responsible for executing the program instructions stored in the memory. The board includes several components such as a USB interface, voltage regulator, crystal oscillator, and power connectors. These components allow the board to operate efficiently and communicate with other electronic devices. The USB interface on the Arduino UNO board allows it to be connected to a computer for programming and data communication. Through the Arduino IDE, users can write the program code, compile it, and upload it directly to the microcontroller. Once the program is uploaded, the Arduino can run independently without the need for continuous computer connection.

The voltage regulator present on the board ensures that the microcontroller receives a stable voltage supply. This helps protect the board from voltage fluctuations and ensures reliable performance during operation. The crystal oscillator provides the clock signal required for the microcontroller to execute instructions at a constant speed.

In the proposed smart factory monitoring system, the Arduino UNO continuously reads input signals from sensors and analyzes them based on the programmed conditions. If abnormal temperature, flame, or vibration levels are detected, the board triggers the alarm system and activates safety mechanisms.[3]



2.1.2 Arduino Uno Description

2.1.3 ESP32 Overview

ESP32 is a powerful and widely used microcontroller module designed for Internet of Things (IoT) applications. It is developed by Espressif Systems and is an advanced version of the ESP8266 Wi-Fi module. The ESP32 provides built-in Wi-Fi and Bluetooth connectivity, which allows devices to communicate with other systems through wireless networks. Because of its high performance, low power consumption, and multiple communication features, ESP32 is widely used in automation, monitoring, and smart device applications.

The ESP32 module contains a dual-core processor which allows it to perform multiple tasks efficiently. It also includes a large number of input and output pins that can be used to connect sensors, actuators, and other electronic components. This flexibility makes ESP32 suitable for building complex embedded systems and IoT-based monitoring solutions.

Another important feature of ESP32 is its ability to connect to the internet directly through Wi-Fi networks. This allows real-time data transmission from sensors to cloud platforms or remote monitoring systems. With this capability, industries and users can monitor devices and systems from anywhere using computers or mobile applications.

In the proposed smart factory monitoring system, ESP32 can be used to transmit sensor data such as temperature, flame detection, and vibration levels to an online monitoring platform. This enables realtime monitoring and helps factory managers observe system conditions remotely. By integrating ESP32 with sensors and microcontrollers, the system becomes more intelligent and efficient for industrial monitoring applications.[2]

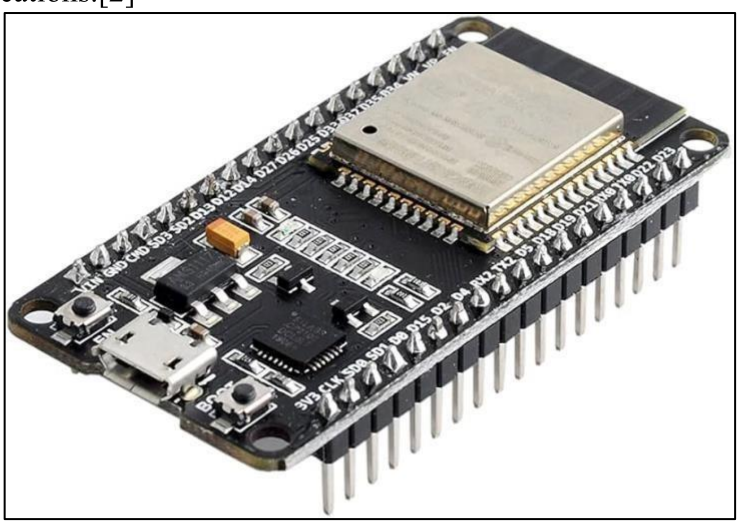


Fig 2.1.3 ESP32 Overview

2.1.4 ESP32 Description

ESP32 is a microcontroller-based development board that combines processing capability with wireless communication technology. It contains a powerful 32-bit processor that operates at a high clock speed, allowing it to process large amounts of data quickly. The board also includes internal memory, analog and digital input/output pins, and communication interfaces such as SPI, I2C, and UART.

The ESP32 module supports both Wi-Fi and Bluetooth connectivity, making it suitable for IoT and wireless communication applications. It can easily connect to a local network and send sensor data to web servers or cloud platforms. This feature enables remote monitoring, data logging, and automation control through internet-based systems.

Another advantage of ESP32 is its energy efficiency. It includes several power-saving modes that allow it to operate with low power consumption, making it suitable for battery-powered devices. The board can also be programmed using the Arduino IDE, which simplifies the development process for beginners and developers.

In this project, ESP32 plays an important role in enabling IoT-based smart factory monitoring. It collects processed sensor data from the microcontroller and sends it through a Wi-Fi network to an online platform. This allows real-time monitoring of factory conditions such as temperature levels, fire detection, and machine vibration. As a result, the ESP32 helps improve industrial safety and enables efficient remote monitoring of the factory environment.

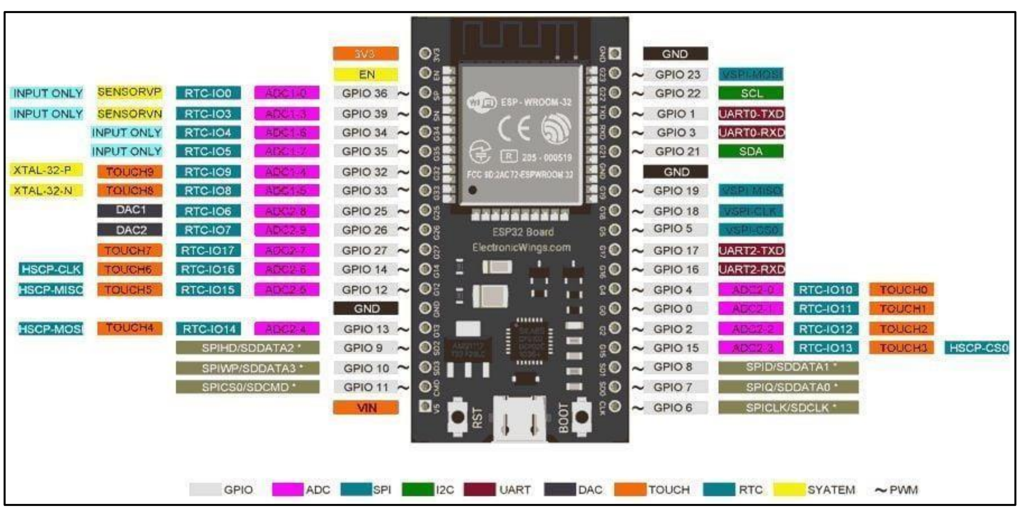


Fig 2.1.4 ESP32 Description

2.1.5 Temperature Sensor

A temperature sensor is an electronic device used to measure and monitor the temperature of a particular environment or object. In industrial environments, machines and electrical equipment generate heat during continuous operation. If the temperature increases beyond the normal operating level, it may damage equipment, reduce machine efficiency, and increase the possibility of fire hazards. Therefore, monitoring temperature levels is an important part of industrial safety and maintenance systems.

In the proposed smart factory monitoring system, the temperature sensor is used to detect the heat levels inside the factory environment and around machinery. The sensor continuously measures the temperature and converts it into an electrical signal which is sent to the Arduino UNO microcontroller. The microcontroller reads the sensor values and processes them according to the program logic.

The microcontroller compares the measured temperature with a predefined threshold value. If the temperature rises above the safe limit, the system identifies it as an abnormal condition. At that moment, the system activates the buzzer to alert workers and operators in the factory so that they can take necessary action immediately.

Continuous monitoring of temperature helps in preventing overheating of machines and reduces the chances of equipment failure. It also improves the safety of the working environment by providing early warning before a serious problem occurs. Temperature sensors are widely used in industrial automation systems because they provide accurate and reliable measurements.

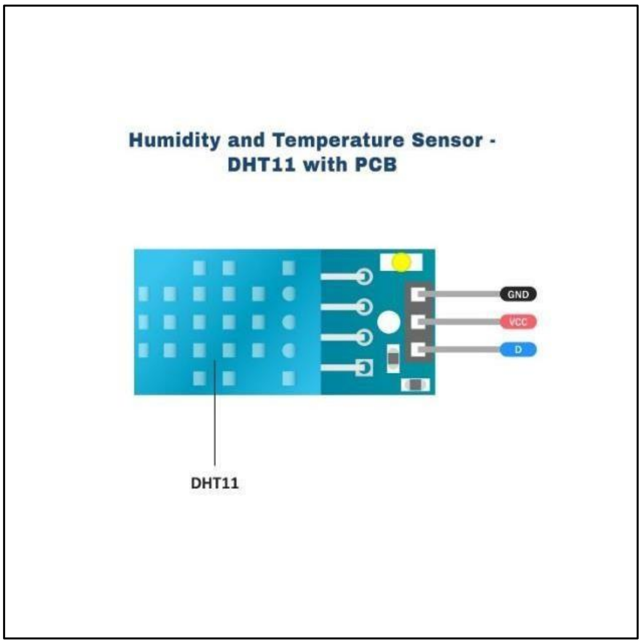


Fig 2.1.5 Temperature sensor

2.1.6 Flame Sensor

The flame sensor is used in this project to detect the presence of fire in the factory environment. Fire accidents in industrial areas can occur due to electrical faults, overheating machines, gas leakage, or chemical reactions. Detecting fire at an early stage is very important to prevent major damage to machinery, infrastructure, and human life. The flame sensor continuously monitors the environment for any signs of flame or fire and provides real-time feedback to the controller.

This sensor works by detecting infrared radiation emitted by flames. Flames produce specific wavelengths of infrared light, and the flame sensor is designed to identify these signals quickly and accurately. When a flame appears within the detection range, the sensor immediately senses the radiation and generates an electrical signal. This signal is then transmitted to the microcontroller for further processing.

After receiving the signal from the flame sensor, the controller analyzes the input and determines whether a fire condition is present. If the controller confirms the presence of a flame, it immediately activates the warning and protection mechanisms integrated into the system. This rapid response capability helps minimize the risk of fire spreading within the factory environment.

Once the fire is detected, the system activates the buzzer to alert workers in the surrounding area. At the same time, the relay is triggered to start the water sprinkler system, which sprays water over the affected area to suppress the flames. By combining flame detection with automatic fire suppression, the system significantly improves safety and reduces the chances of severe fire damage in the smart factory.[8]

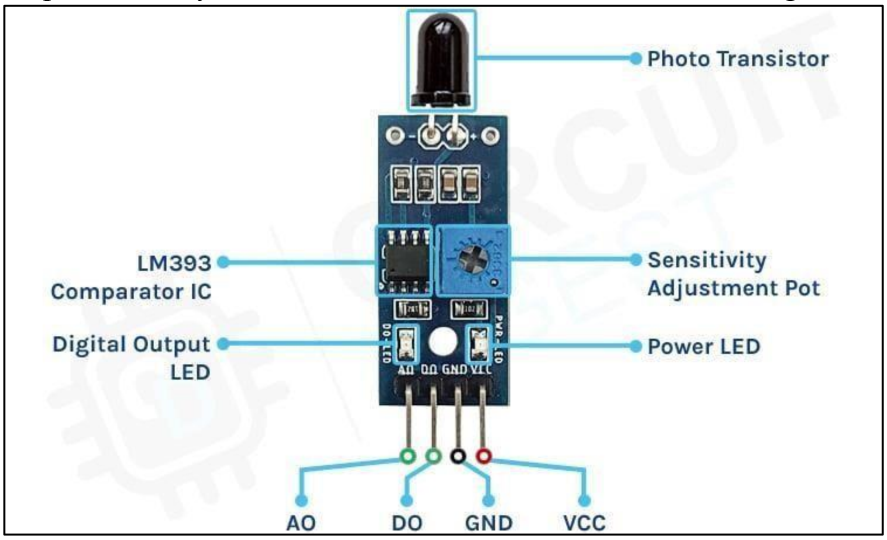


Fig 2.1.6 Flame Sensor

2.1.7 Piezo Electric Vibration Sensor

The vibration sensor is used in this project to monitor mechanical vibrations produced by industrial machines. In a smart factory environment, machines operate continuously and may experience vibrations due to moving parts, mechanical stress, or operational loads. While normal vibrations are expected during machine operation, excessive vibrations may indicate mechanical faults, imbalance, or wear and tear of machine components.

Continuous monitoring of vibration levels is important for maintaining machine health and preventing unexpected failures. The vibration sensor detects movement or oscillations in machines and converts them into electrical signals that can be analyzed by the controller. By monitoring these signals, the system can identify abnormal vibration patterns that may indicate potential problems.

Mechanical faults such as loose components, misalignment, bearing damage, or motor imbalance often cause unusual vibrations in industrial machines. If these issues are not detected early, they may lead to serious damage, machine breakdown, or costly maintenance. The vibration sensor helps in detecting such conditions at an early stage, allowing maintenance teams to take preventive actions.

When the vibration level exceeds the predefined safe threshold, the controller recognizes it as a fault condition. The system then activates the buzzer to alert workers about the abnormal machine condition. This early warning system helps prevent equipment failure, reduces maintenance costs, and improves overall operational efficiency in the smart factory system.[7]

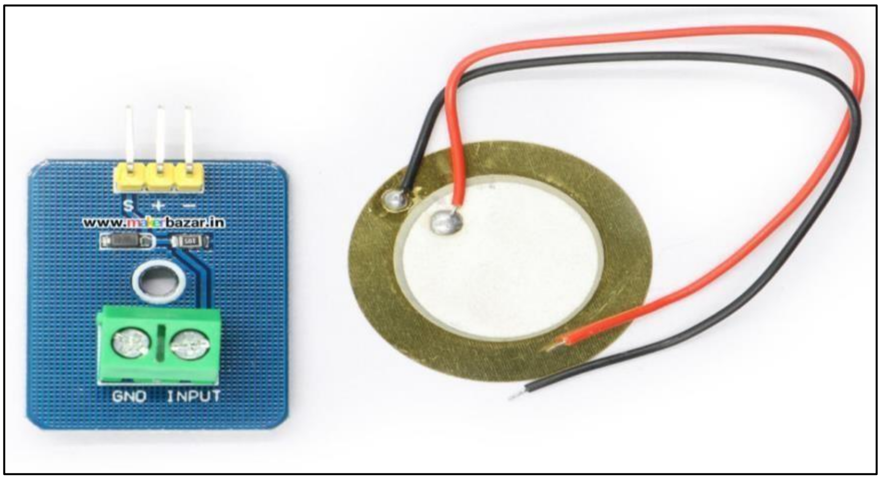


Fig 2.1.7 Vibration sensor

2.1.8 Relay and Buzzer

The relay and buzzer are important output components used in this project to provide alerts and control external devices. The relay acts as an electrically operated switch that allows the microcontroller to control high-power devices using low-power electrical signals. This feature is essential because the controller itself cannot directly operate high-power equipment such as water sprinklers or industrial devices.

When the controller receives signals from sensors indicating abnormal conditions such as high temperature, fire detection, or excessive vibration, it processes the data and decides the necessary action. The relay is then activated by the controller to switch on external safety devices. This allows the system to automatically respond to dangerous situations without human intervention.

The buzzer is used as an audible alarm device to warn workers and operators about emergency situations in the factory. When activated, the buzzer produces a loud sound that immediately alerts people nearby. This quick notification allows workers to respond to the situation and take appropriate safety measures.

The combination of relay and buzzer ensures that the system not only detects faults but also provides immediate alerts and activates safety mechanisms. This improves the effectiveness of the smart factory monitoring system and ensures better protection for machines, workers, and factory infrastructure.

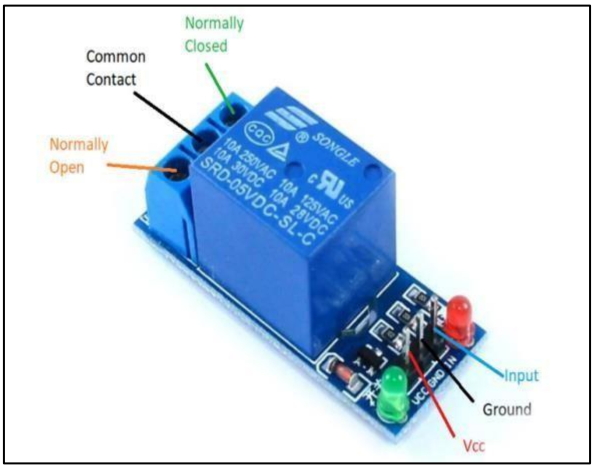


Fig 2.1.8 Relay and Buzzer

2.1.9 Water Sprinkler

The water sprinkler system is used in this project as an automatic fire suppression mechanism. In industrial environments, fire can spread rapidly and cause severe damage if not controlled quickly. The water sprinkler plays a vital role in controlling fire at an early stage and minimizing potential losses.

When the system detects abnormal conditions such as high temperature or the presence of flame, the controller immediately processes the sensor data and activates the relay. The relay then switches on the water sprinkler system, allowing water to be sprayed over the affected area. This helps reduce the temperature and control the spread of fire.

The sprinkler system is designed to respond automatically without requiring manual intervention. This is especially important in large factories where quick human response may not always be possible. The automatic activation of the sprinkler ensures that fire suppression begins immediately after detection.

By integrating the water sprinkler with temperature and flame sensors, the project provides a complete fire protection solution for smart factories. This system improves workplace safety, protects valuable equipment, and reduces the risk of major fire accidents in industrial environments.

Fig 2.1.9 Water Sprinkler

2.2 SOFTWARE INSTALLATION

Arduino – Installation

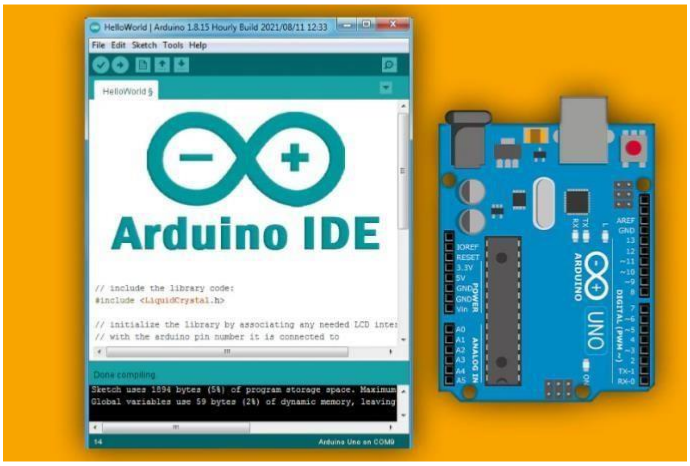


Fig 2.2.1: **Arduino IDE and Arduino UNO Board**

After learning about the main parts of the Arduino UNO board, we are ready to learn how to set up the Arduino IDE. Once we learn this, we will be ready to upload our program on the Arduino board. In this section, we will learn in easy steps, how to set up the Arduino IDE on our computer and prepare the board to receive the program via USB cable.

Step 1: First you must have your Arduino board (you can choose your favourite board) and a USB cable. In case you use Arduino UNO, Arduino Duemilanove, Nano, Arduino Mega 2560, or Diecimila, you will need a standard USB cable (A plug to B plug), the kind you would connect to a USB printer as shown in the following image. [9]

Fig 2.2.2: USB cable

In case you use Arduino Nano, you will need an A to Mini-B cable instead as shown in the following image.

Step 2: Download Arduino IDE Software:

You can get different versions of Arduino IDE from the Download page on the Arduino Official website. You must select your software, which is compatible with your operating system (Windows, IOS, or Linux). After your file download is complete, unzip the file.

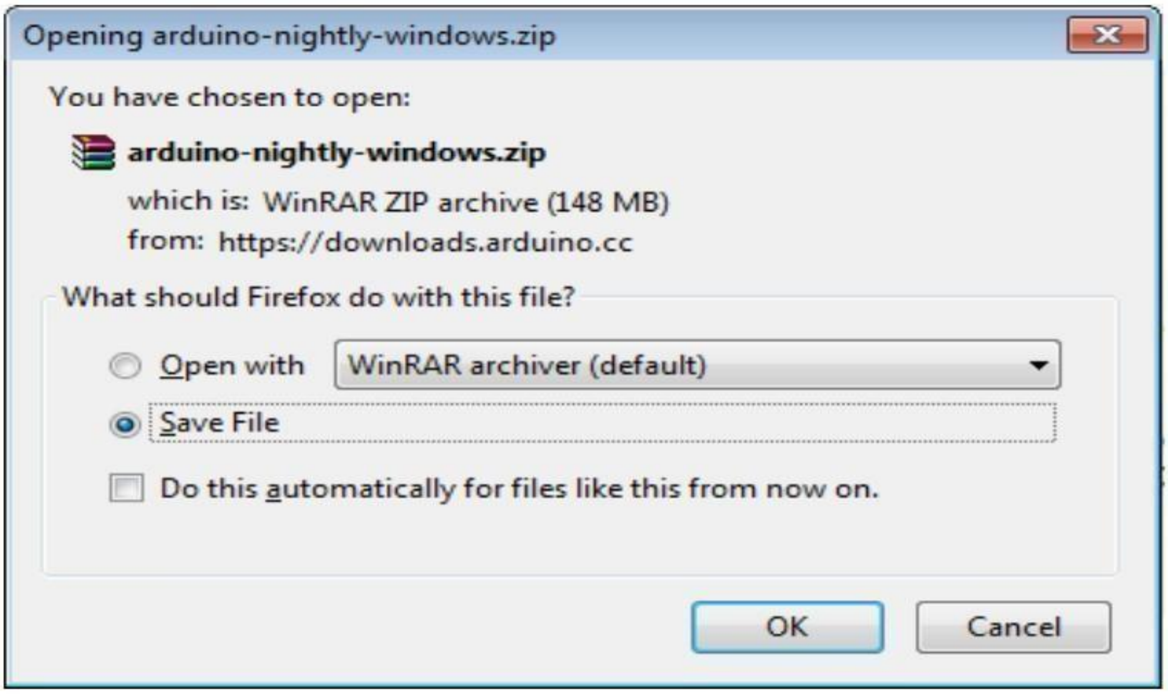


Figure 2.2.3 Download Arduino IDE Software

Step 3: Power up your board:

The Arduino Uno, Mega, Duemilanove and Arduino Nano automatically draw power from either, the USB connection to the computer or an external power supply. If you are using an Arduino Diecimila, you have to make sure that the board is configured to draw power from the USB connection. The power source is selected with a jumper, a small piece of plastic that fits onto two of the three pins between the USB and power jacks. Check that it is on the two pins closest to the USB port. Connect the Arduino board to your computer using the USB cable. The green power LED (labelled PWR) should glow.

Step 4: Launch Arduino IDE:

After your Arduino IDE software is downloaded, you need to unzip the folder. Inside the folder, you can find the application icon with an infinity label (application.exe). Double-click the icon to start the IDE.

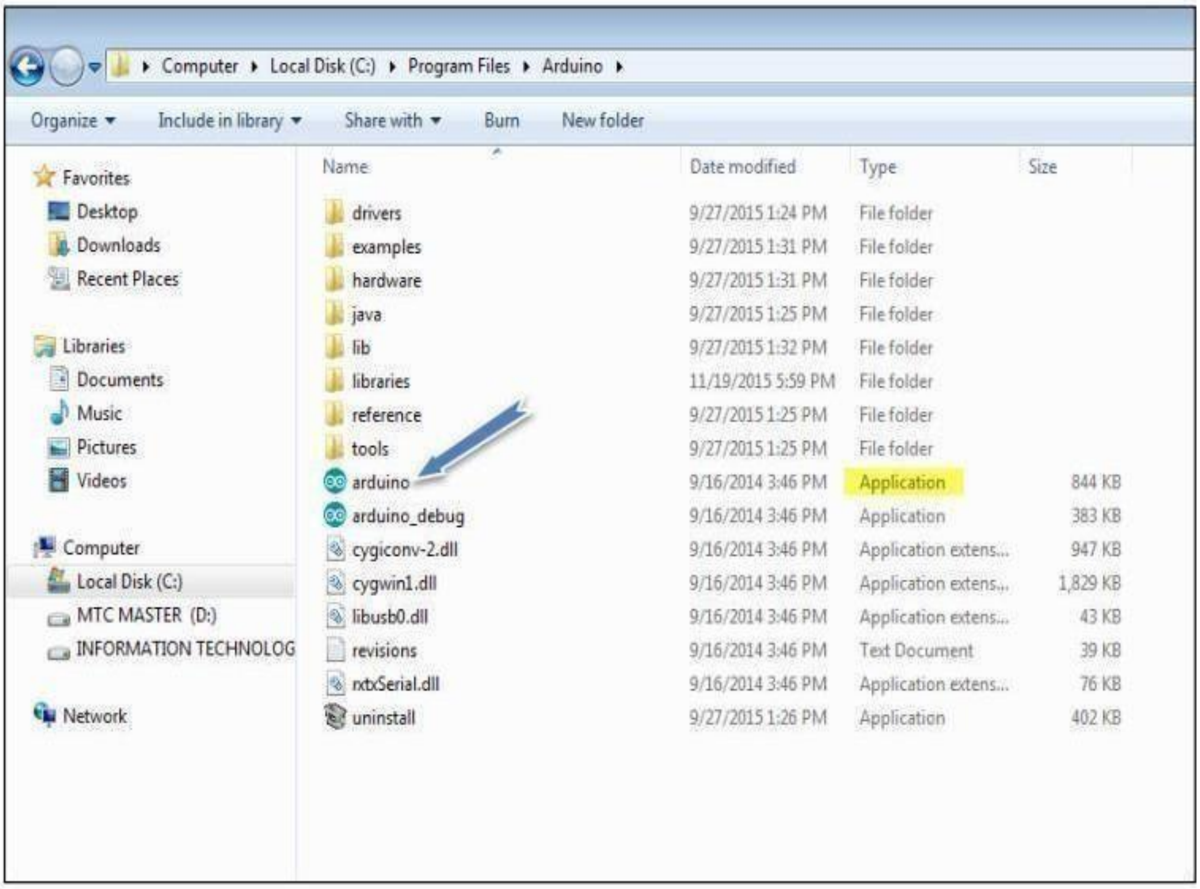


Figure 2.2.4 Launch Arduino IDE

Step 5: Open your first project:

Once the software starts, you have two options:

- Create a new project.
- Open an existing project example. To create a new project, select

  File --> New.

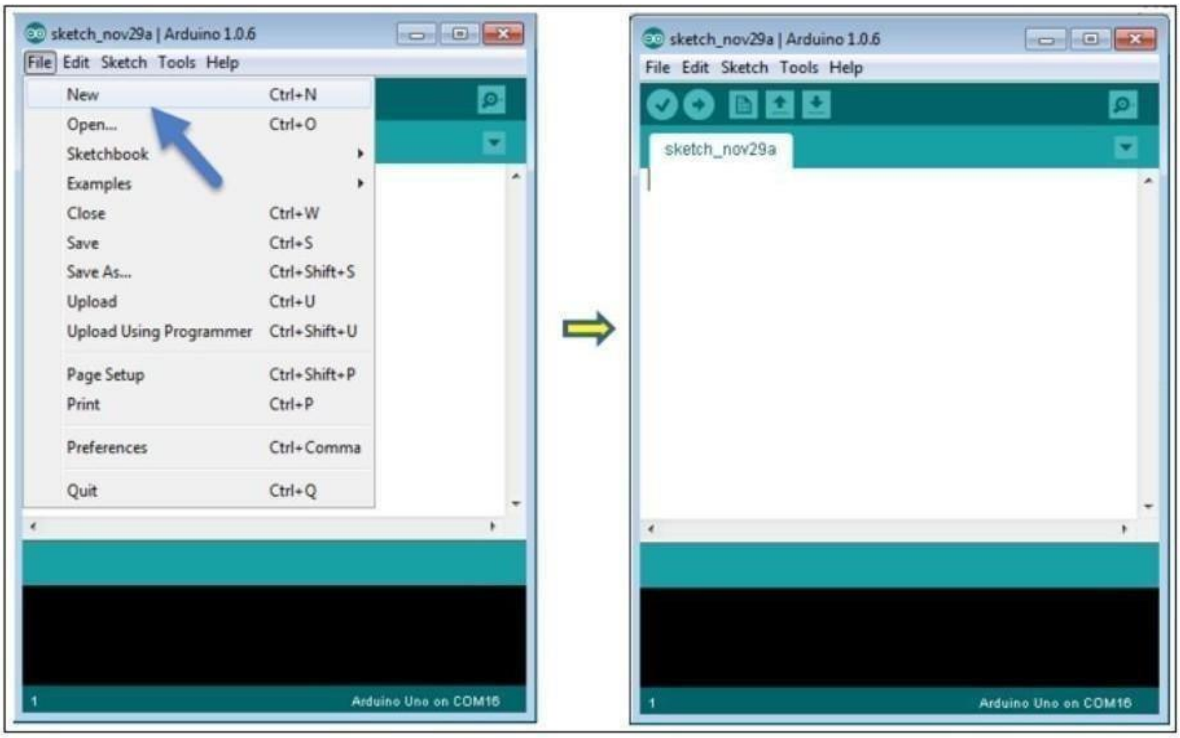


Figure 2.2.5 To open New Project

To open an existing project example, select File -> Example -> Basics -> Blink.

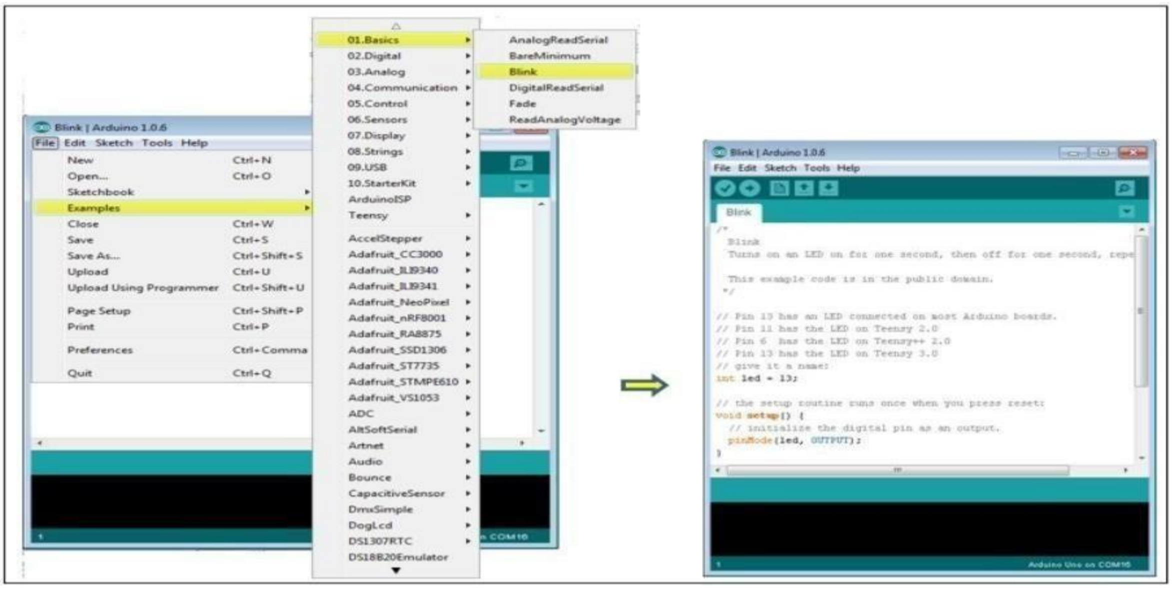


Figure 2.2.6 To open an existing project example

Here, we are selecting just one of the examples with the name Blink. It turns the LED on and off with some time delay. You can select any other example from the list.

Step 6: Select your Arduino board:

To avoid any error while uploading your program to the board, you must select the correct Arduino board name, which matches with the board connected to your computer. Go to Tools -> Board and select your board.

ESP32 INSTALLATION

The ESP32 is a powerful microcontroller developed by Espressif Systems that is widely used in IoT and embedded system applications. It has built-in Wi-Fi and Bluetooth capabilities, which makes it suitable for wireless communication and smart automation projects. Similar to Arduino boards, the ESP32 can also be programmed using the Arduino Integrated Development Environment (Arduino IDE). This allows developers and students to write programs using simple C/C++ syntax and upload them easily to the ESP32 board.

Programming the ESP32 using Arduino IDE is very convenient because the environment provides a user-friendly interface, built-in libraries, and example programs. With the help of Arduino IDE, users can write code, compile it, and upload it directly to the ESP32 board through a USB cable. This makes it easier to develop applications such as IoT devices, smart home systems, sensor monitoring systems, and automation projects.

To program ESP32 using Arduino IDE, first the ESP32 board support package must be installed. This can be done by adding the ESP32 board manager URL in the Arduino IDE preferences and then installing the ESP32 boards from the Board Manager. After installation, the ESP32 board can be selected from the Tools → Board menu.

Once the board is selected, the user can connect the ESP32 to the computer using a USB cable. The correct COM port must also be selected from the Tools menu. After this setup, the user can write the program in the Arduino IDE editor window. The code is written in a simplified version of C/C++ language, similar to Arduino programming.

After writing the program, the Verify button is used to compile the code and check for errors. If the compilation is successful, the Upload button is used to transfer the program to the ESP32 board. The microcontroller then executes the uploaded code and performs the desired operations such as reading sensor data, controlling motors, sending data over Wi-Fi, or communicating with other devices.

The ESP32 supports both digital and analog input/output pins, PWM signals, serial communication, and various communication protocols such as I2C, SPI, and UART. These features make it highly suitable for advanced embedded applications. Additionally, the availability of numerous open-source libraries and community support makes ESP32 development easier for beginners as well as professionals.

In this project, the ESP32 can also be programmed using Arduino IDE to read sensor values and control actuators similar to the Arduino Uno. The flexibility and wireless capability of ESP32 allow future expansion of the system into an IoT-based crop monitoring and protection system where sensor data can be monitored remotely.

Thus, Arduino IDE provides a simple and efficient platform for developing and uploading programs to the ESP32 microcontroller, making it a powerful tool for embedded system and IoT applications.[10]

Fig 2.2.7 ESP32 in Arduino ide

2.2.7 PROGRAM

Arduino Uno Code

```
#include <DHT.h>

#define DHTPIN 2
#define DHTTYPE DHT11

DHT dht(DHTPIN, DHTTYPE);

int flame1 = 3; int
flame2 = 4; int
vibration = A0;

int relay = 8; int
buzzer = 9;

void setup() {
Serial.begin(9600);

pinMode(flame1, INPUT);
pinMode(flame2, INPUT);
pinMode(vibration, INPUT);

pinMode(relay,OUTPUT);
 pinMode(buzzer, OUTPUT);

digitalWrite(relay,LOW);
 digitalWrite(buzzer, LOW);

dht.begin();
}

void loop()
{

float temp = dht.readTemperature();

int flameState1 = digitalRead(flame1); int
flameState2 = digitalRead(flame2);

int vibrationValue = analogRead(vibration);

Serial.print(temp);
Serial.print(",");
```

```
Serial.print(flameState1); Serial.print(",");
Serial.print(flameState2);
Serial.print(",");
Serial.println(vibrationValue);

if(flameState1 == LOW || flameState2 == LOW)
{ digitalWrite(relay, HIGH);  // sprinkler
ON digitalWrite(buzzer, HIGH); // alarm
}

else if(vibrationValue > 500)
{  digitalWrite(buzzer,  HIGH);    // vibration
alert digitalWrite(relay, LOW);
}

else { digitalWrite(relay,
LOW);
digitalWrite(buzzer,
LOW);
}

delay(2000);
}
```

ESP32 Code

```
#include <ESP8266WiFi.h>

const char* ssid = "YOUR_WIFI_NAME"; const char*
password = "YOUR_WIFI_PASSWORD";

WiFiServer server(80);

void setup()
{
Serial.begin(9600);

WiFi.begin(ssid, password);

while(WiFi.status() != WL_CONNECTED)
{ delay(500);
}

server.begin();
}
```

```
void loop()
{

WiFiClient client = server.available();

if(client)
{

String data = Serial.readStringUntil('\n');

client.println("HTTP/1.1 200 OK");
client.println("Content-Type: text/html");
client.println("");

client.println("<html><body>");
client.println("<h2>Smart Factory
Monitoring</h2>"); client.println("<p>Sensor
Data:</p>"); client.println(data);
client.println("</body></html>");

delay(1000);

client.stop();

}
```

CHAPTER 3

PROJECT DESCRIPTION

3.1 Block Diagram

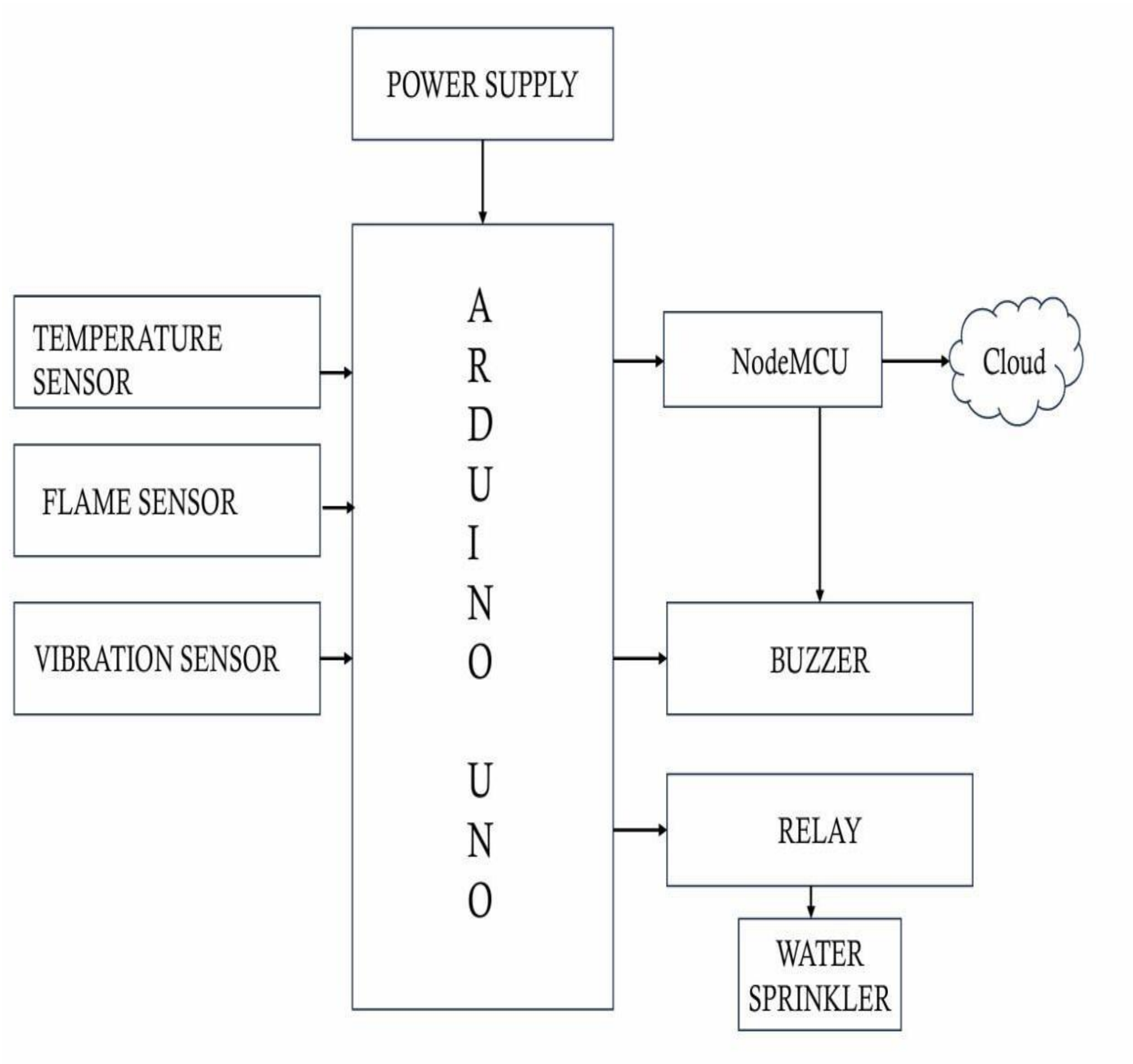


Fig 3.1: Basic Block Diagram

3.2 Circuit Diagram

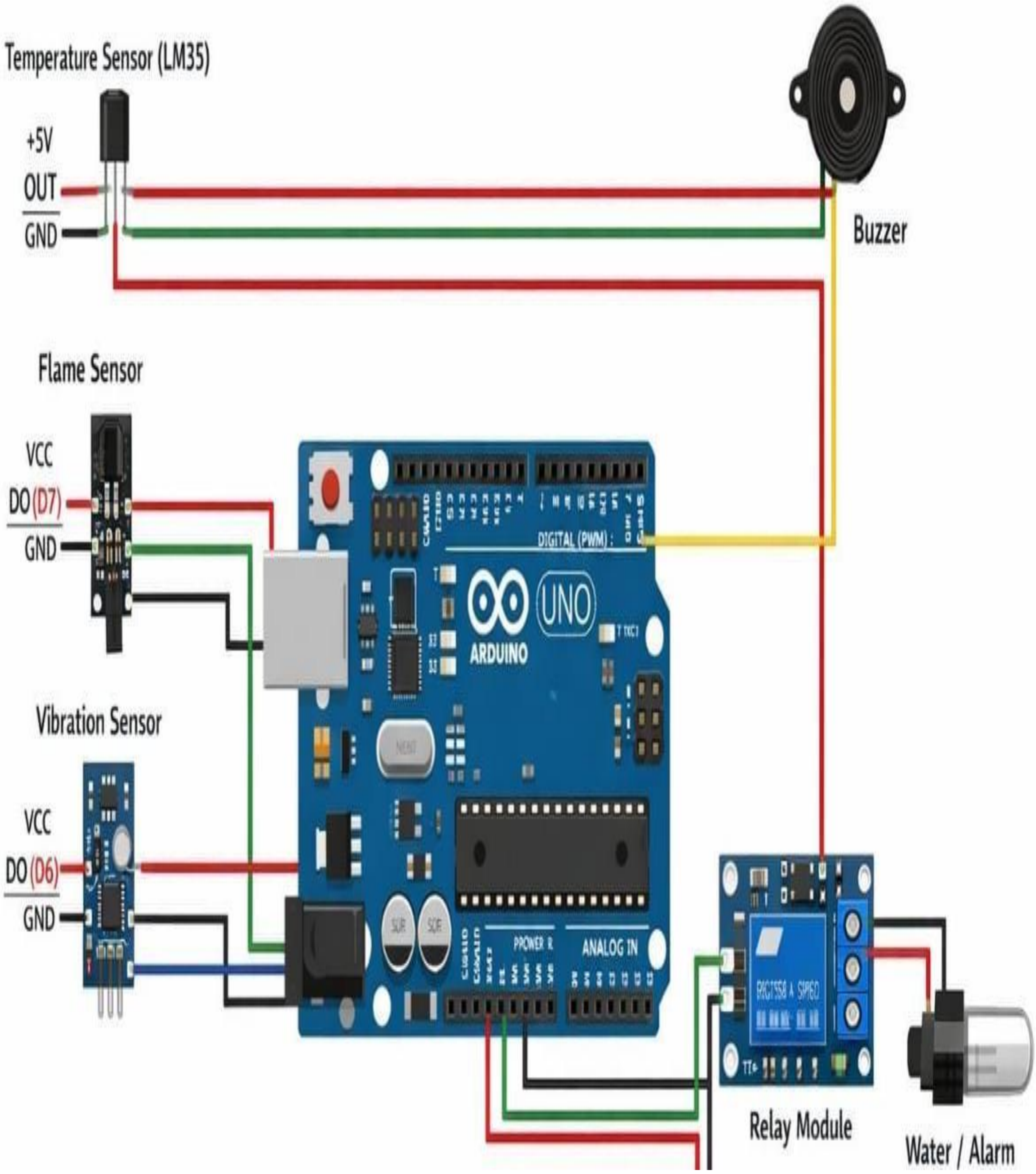


Fig 3.2: Circuit Diagram

3.2.5 Working Principal

The Real Time Smart Factory Management and Fault Detection System using Temperature, Flame and Vibration Sensors is designed to improve safety and reliability in industrial environments. In modern industries, machines operate continuously for long hours, and any abnormal condition such as overheating, fire hazards, or excessive vibration may cause serious damage to equipment and may also lead to dangerous accidents. Therefore, it is important to continuously monitor factory conditions and detect faults at an early stage. This system achieves that objective by using multiple sensors such as the DHT11 temperature sensor, flame sensor, and piezoelectric vibration sensor along with a microcontroller-based monitoring system.

The system mainly consists of sensors, a microcontroller unit, communication modules, and output devices. The sensors are responsible for collecting real-time data from the factory environment and machines. The collected data is then sent to the Arduino UNO microcontroller, which acts as the central processing unit of the system. The microcontroller processes the sensor signals and determines whether the system is operating under normal conditions or if any abnormal situation has occurred. If the system detects a fault, it immediately activates alert and safety mechanisms such as a buzzer alarm and a water sprinkler system.

The DHT11 temperature sensor plays an important role in monitoring the temperature of the industrial environment. Machines and electrical equipment generate heat during continuous operation, and excessive temperature may lead to machine failure or fire hazards. The DHT11 sensor continuously measures the temperature and sends the digital data to the Arduino UNO microcontroller. The controller reads the temperature value and compares it with a predefined threshold limit programmed in the system. If the measured temperature exceeds the safe limit, the controller interprets this as an overheating condition. In response to this condition, the system activates the buzzer to alert workers and operators in the factory so that necessary actions can be taken to prevent equipment damage or accidents.[4]

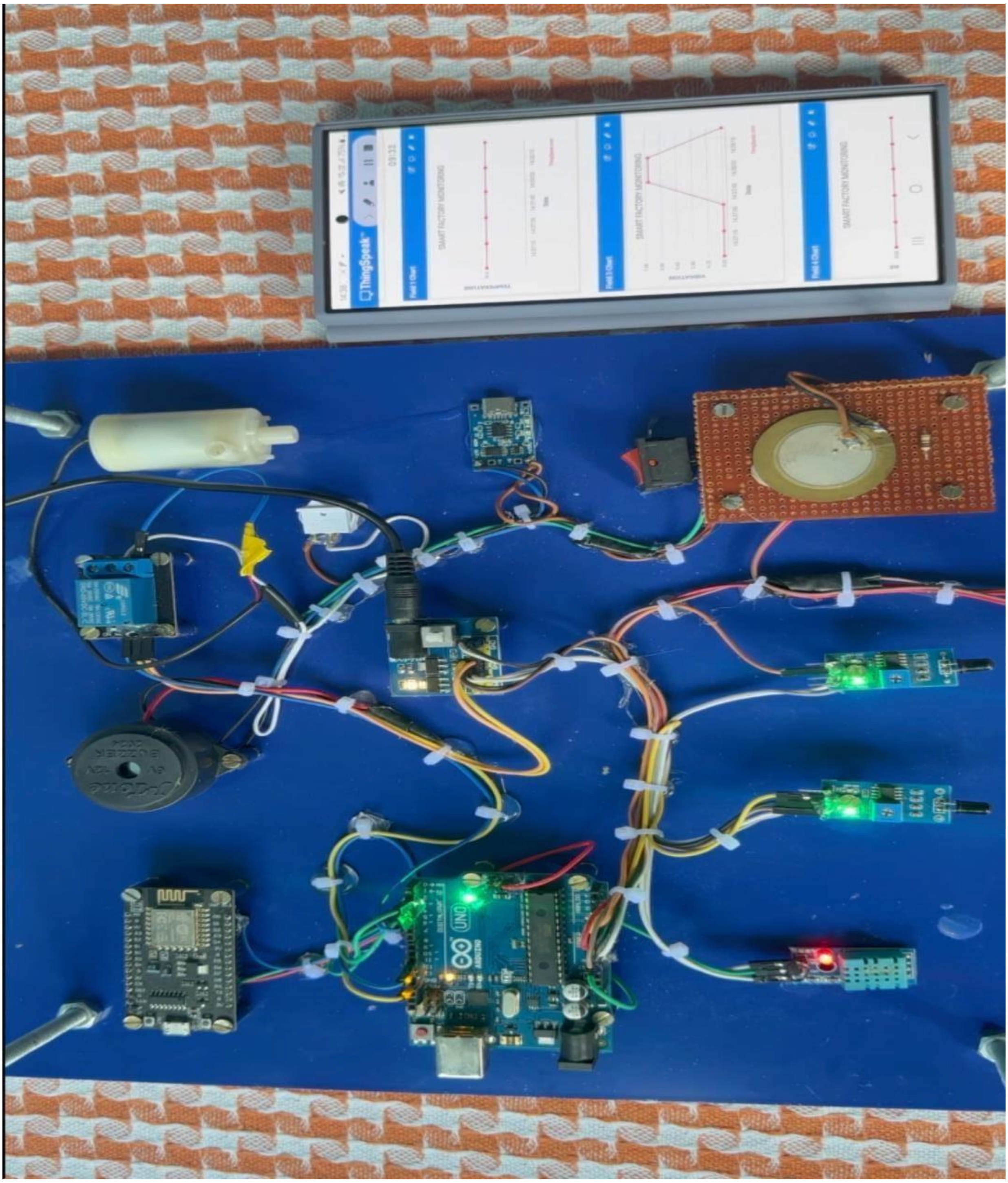
Another important component used in the system is the flame sensor, which is responsible for detecting fire in the factory environment. Fire accidents are one of the most dangerous hazards in industrial areas because they can spread quickly and cause severe damage to machinery and infrastructure. The flame sensor works by detecting infrared radiation emitted by flames. When a flame is present within the sensor's detection range, it senses the infrared radiation and generates a signal that is sent to the Arduino UNO microcontroller. After receiving the signal from the flame sensor, the controller immediately processes the information and confirms the presence of fire. Once fire is detected, the controller activates the relay module, which turns on the water sprinkler system. The sprinkler sprays water over the affected area to control the fire and prevent further spreading. At the same time, the buzzer alarm is activated to warn

workers about the fire hazard and help them respond quickly to the emergency situation.
The piezoelectric vibration sensor is used to monitor vibrations produced by industrial machines. In factories, machines often produce vibrations due to moving mechanical parts, motors, and rotating components. While normal vibrations are expected during machine operation, excessive or abnormal vibrations may indicate mechanical faults such as loose components, misalignment, imbalance, or worn-out bearings. The vibration sensor continuously monitors the vibration levels of the machines and converts the mechanical vibrations into electrical signals. These signals are sent to the Arduino UNO for analysis. The controller reads the vibration values and compares them with the predefined threshold levels programmed in the system. If the vibration level exceeds the acceptable range, the controller identifies it as an abnormal condition. In this situation, the buzzer is activated to notify workers or maintenance personnel about the machine fault so that necessary inspection and maintenance can be performed before serious damage occurs.
In addition to the sensors and microcontroller, the system also includes output devices such as a buzzer, relay module, and water sprinkler. The buzzer acts as an audible warning system that alerts workers and operators whenever an abnormal condition is detected in the factory. The relay module acts as an electrically controlled switch that allows the microcontroller to control high-power devices such as the water sprinkler system. Since the microcontroller cannot directly operate high-power devices, the relay acts as an interface between the microcontroller and external safety devices. When the controller detects a fire condition, it sends a signal to the relay module, which then activates the sprinkler system to spray water and control the fire.
The NodeMCU module is used in the system to provide Internet of Things (IoT) connectivity. The NodeMCU receives sensor data from the Arduino UNO through serial communication and transmits the data through Wi-Fi to a remote monitoring platform or web server. This allows factory managers and operators to monitor the system conditions from anywhere using computers or mobile devices connected to the internet. Real-time monitoring enables quick decision-making and allows operators to take immediate action when abnormal conditions are detected.
The entire system works continuously to monitor the industrial environment and machine conditions. All sensors operate simultaneously and send real-time data to the microcontroller. The Arduino UNO continuously reads and processes the data from the temperature sensor, flame sensor, and vibration sensor. If any sensor detects an abnormal condition such as high temperature, fire detection, or excessive vibration, the controller immediately activates the alarm system and safety mechanisms. This automatic response helps reduce the risk of accidents and prevents major equipment damage.

Another advantage of this system is that it reduces the need for constant manual supervision. In traditional industrial monitoring systems, workers must regularly inspect machines and factory conditions to identify possible faults. However, manual monitoring is not always reliable because faults can occur suddenly between inspection periods. By implementing an automated monitoring system with sensors and microcontrollers, the system can detect faults immediately and alert operators without delay. This improves safety and ensures continuous and efficient factory operations. Overall, the Real Time Smart Factory Management and Fault Detection System provides an effective solution for monitoring industrial environments. By integrating sensors, microcontrollers, IoT communication modules, and automatic safety devices, the system ensures early detection of faults and rapid response to dangerous situations. The system helps prevent accidents such as machine overheating, fire hazards, and mechanical failures. It also improves industrial productivity by reducing unexpected machine breakdowns and maintenance costs.

Thus, the proposed system demonstrates how modern embedded systems and sensor technologies can be used to build intelligent monitoring solutions for smart factories. By continuously observing critical parameters such as temperature, flame detection, and machine vibration, the system improves safety, reliability, and efficiency in industrial environments. This project highlights the importance of automation and IoT technology in developing advanced smart factory systems that can operate safely and efficiently in modern industrial applications.

CHAPTER 4
RESULTS



CHAPTER 5
ADVANTAGES, DISADVANTAGES, AND APPLICATIONS

5.1 Advantages

The advantages of Real Time Smart Factory Management and Fault Detection System Using Temperature, Flame and Vibration are numerous and impactful. Here are some key benefits:

- Improves Industrial Safety:
- The system continuously monitors factory conditions such as temperature, flame, and vibration. Early detection of abnormal conditions helps prevent accidents like fire hazards and machine failures
- Early fault detection:
- Sensors detect abnormal machine behaviour at an early stage, allowing operators to take preventative actions before serious damage occurs
- Real-Time Monitoring:
- The system provides real-time monitoring of machine conditions, which helps in quick identification of faults and ensures smooth industrial operations.
- Reduces Maintenance Cost:
- By detecting machine faults early, the system reduces expensive repair costs and increases the lifespan of industrial equipment.
- Automatic Alert System:
- When abnormal conditions are detected, the system activates alarms such as a buzzer or warning indicator to alert the factory operators.
- Improves Productivity:
- Continuous monitoring helps reduce unexpected machine breakdowns, which increases overall production efficiency.
- Easy to Implement:
- The system can be easily implemented using a microcontroller like the Arduino Uno along with simple sensors
- Low Cost System:
- The sensors used in the project are inexpensive and easily available, making the system affordable for industries.

5.2 Disadvantages

While Real Time Smart Factory Management and Fault Detection System Using Temperature, Flame and Vibration many advantages, there are also some disadvantages to consider:

- Limited Sensor Range:
- Sensors may only detect faults within a limited range, which may not cover the entire factory area.
- Initial Setup Cost:
- Although components are inexpensive, the initial setup and installation may require some investment
- Sensor Sensitivity Issues:
- Sometimes sensors may produce false readings due to environmental noise or interference.
- Requires Power Supply:
- The system needs a continuous power supply for proper functioning.
- Maintenance Required:
- Sensors and electronic components require periodic maintenance and calibration.
- Limited Fault Detection:
- The system can only detect faults related to temperature, flame, and vibration, not all types of machine problems.
- Dependency on Microcontroller:
- If the controller fails, the entire monitoring system may stop functioning.
- Environmental Effects:
- Dust, humidity, or extreme temperatures in factories may affect sensor performance.
- Wiring Complexity:
- Multiple sensors and modules require proper wiring connections, which may increase installation complexity.
- Limited Data Storage:
- Basic microcontroller systems may not store large amounts of data without additional memory.

5.3 Applications

Real Time Smart Factory Management and Fault Detection System Using Temperature, Flame and Vibration has various applications that can significantly enhance agricultural practices. Here are some key applications:

- Smart Manufacturing Industries:
    - Used to monitor machine conditions and improve safety in automated factories.
- Industrial Machine Monitoring:
    - Helps detect machine faults in manufacturing plants and production units.
- Fire Detection Systems:
    - Flame sensors can detect fire hazards in industrial environments.
- Power Plants:
    - Used to monitor equipment temperature and vibration levels in power generation units.
- Chemical Industries:
    - Helps detect overheating or fire risks in chemical processing plants.
- Automobile Manufacturing Plants:
    - Used for monitoring machinery used in automobile production lines.
- Oil and Gas Industries:
    - Helps detect abnormal machine vibrations and fire hazards.
- Warehouse Safety Systems:
    - Used to detect fire or overheating in storage areas.
- Smart Factory Automation:
    - Forms a part of Industry 4.0 systems for intelligent factory monitoring.
- Predictive Maintenance Systems:
    - Helps industries predict machine failures and perform maintenance before breakdown occurs.

CHAPTER 6
CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

The Real-Time Smart Factory Management and Fault Detection System is designed to improve safety and efficiency in industrial environments. In this project, different sensors such as a temperature sensor, flame sensor, and vibration sensor are used to continuously monitor the working conditions of machines and the surrounding environment. The sensors are connected to the Arduino UNO, which processes the data and detects abnormal conditions such as high temperature, fire, or unusual machine vibration.

When any unsafe condition is detected, the system immediately activates warning and protection mechanisms. The buzzer provides an audible alert to notify workers about the problem. In case of fire detection, the relay module activates the water sprinkler, which helps in controlling the fire quickly and preventing further damage. At the same time, the NodeMCU module enables wireless communication and allows the system to send monitoring data through the internet for remote supervision.

This project demonstrates how Internet of Things (IoT) technology can be used to build a smart monitoring system for factories. It helps in reducing manual monitoring, increasing safety, and preventing major accidents. The system is cost-effective, reliable, and easy to implement in small or medium-scale industries. Overall, the developed system shows how modern embedded systems and IoT technologies can contribute to building smart, safe, and automated industrial environments.

In addition, this project helps students understand the practical implementation of embedded systems, sensor interfacing, and IoT communication in real-world industrial applications. It provides valuable knowledge about how modern technology can be used to detect faults early and prevent accidents in factories. By combining hardware components with software programming, the system demonstrates an effective and reliable solution for industrial monitoring. Therefore, this project plays an important role in promoting automation, safety, and smart management in modern industries.

6.2 Future Scope:

The future scope of Real Time Smart Factory Monitoring And Fault Detection Using Temperature , Flame And Vibration is quite promising and can be explored in several ways:

The proposed system has many possibilities for improvement and expansion in the future. One of the major improvements can be the integration of additional sensors such as gas sensors, humidity sensors, smoke detectors, and pressure sensors to monitor more environmental conditions inside the factory. This will make the system more accurate and capable of detecting a wider range of industrial hazards.

Another future enhancement is the development of a mobile application or web dashboard where factory managers can monitor real-time data, receive instant notifications, and control safety devices remotely. This would make the system more user-friendly and suitable for large industrial environments.

The project can also be upgraded by implementing Artificial Intelligence (AI) or Machine Learning algorithms to analyze sensor data and predict machine failures before they occur. Predictive maintenance can help industries reduce downtime, avoid expensive repairs, and improve overall productivity.

In addition, the system can be expanded to support multiple machines and multiple factory sections by using advanced IoT communication networks. Integration with cloud platforms and data analytics tools can also help in storing long-term data and generating reports for better decision making.

Thus, with further development and integration of advanced technologies, this system can evolve into a complete smart factory monitoring and automation solution, making industrial operations safer, more efficient, and highly intelligent.

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